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Lema de cambio de variable para el laplaciano

Lema 1 (Fórmulas de cambio de variable para el laplaciano). *Sea $f : \Omega_1 \to \Omega_2$ una función holomorfa y sea $\phi : \Omega_2 \to \mathbb{R}^n$ una función $\mathcal{C}^2$*

$$\implies \forall z \in \Omega_1 : \Delta\big(\phi \circ f\big)(z) = |f'(z)|^2 \left(\Delta\phi\right)\big(f(z)\big).$$

***Demostración*:** Sea $w = f(z)$ la variable en Ω_2. Por la regla de la cadena para las derivadas de Wirtinger, tenemos que

$$\frac{\partial}{\partial z}\frac{\partial}{\partial \overline{z}}\left(\phi \circ f\right)(z) = \frac{\partial}{\partial z}\left(\frac{\partial \phi}{\partial w}(f(z))\,\frac{\partial f}{\partial \overline{z}}(z) + \frac{\partial \phi}{\partial \overline{w}}(f(z))\,\frac{\partial \overline{f}}{\partial \overline{z}}(z)\right)$$

Como f es holomorfa, por la `prop-fn-holomorfa-iff-wirtinger/??`, $\frac{\partial f}{\partial \overline{z}} \equiv 0$ y el primer sumando se anula. Luego, por la regla del producto, tenemos que

$$\frac{\partial}{\partial z}\frac{\partial}{\partial \overline{z}}\left(\phi \circ f\right)(z) = \frac{\partial}{\partial z}\left(\frac{\partial \phi}{\partial \overline{w}} \circ f\right)(z)\,\frac{\partial \overline{f}}{\partial \overline{z}}(z) + \frac{\partial \phi}{\partial \overline{w}}(f(z))\,\frac{\partial}{\partial z}\frac{\partial \overline{f}}{\partial \overline{z}}(z).$$

Como f es holomorfa, $\overline{f}$ es antiholomorfa y, por tanto, $\frac{\partial}{\partial z}\frac{\partial \overline{f}}{\partial \overline{z}} \equiv 0$ y el segundo sumando se anula. Aplicando de nuevo la regla de la cadena y usando que f es holomorfa para aplicar el `cor-wirtinger-composicion-holomorfas/`Corolario 1, tenemos que

$$\frac{\partial}{\partial z}\frac{\partial}{\partial \overline{z}}\left(\phi \circ f\right)(z) = \frac{\partial^2 \phi}{\partial w\,\partial \overline{w}}(f(z))\,\frac{\partial f}{\partial z}(z)\,\frac{\partial \overline{f}}{\partial \overline{z}}(z) = \frac{\partial^2 \phi}{\partial w\,\partial \overline{w}}(f(z))\,\left|f'(z)\right|^2.$$

Entonces, por el `lem-laplaciano-wirtinger/??`, tenemos que $\Delta\left(\phi \circ f\right)(z) = \left|f'(z)\right|^2\,\left(\Delta\phi\right)\left(f(z)\right)$. ■

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